

Interdisciplinary Centre for Water Research invites applications for 2 RA (Postdoc) in an international collaborative project funded by Schmidt Science

We are looking to hire two enthusiastic researchers in a cutting edge international project: “Unlocking local knowledge production for global water reanalysis” led by the Imperial college London and International Water Management Institute. The project is funded by the Schmidt Science foundation and partner institutes are: University of Cyprus (UCyprus), University of Cuenca (UCuenca), Pontifical Catholic, University of Peru (PUCP), Swiss Federal Technology Institute of Lausanne (EPFL), University of Ghana (UGhana) and ATUK Strategic Consulting (ATUK).

General overview of the project: Global freshwater systems are critically threatened by environmental change and over-exploitation, stressing the need for novel, transformative solutions. Because of the diversity and complexity of social-hydrological systems, such solutions need to be informed by solid scientific evidence as well as tailored to local realities, knowledge, and practices. The current generation of reanalysis systems struggle to produce evidence that is accurate, tailored, and actionable at the local scale. We will develop a transformational data assimilation framework to integrate local knowledge together with existing and other emerging data sources, in global water system models. We will focus on three knowledge sources that are currently underrepresented in global models: nonstatutory monitoring, citizen observations, and local knowledge. Leveraging state-of-the-art technologies such as semantic data models and machine learning, we will develop novel knowledge integration workflows, with explicit tracking of epistemic and aleatoric uncertainties. Using a flexible modelling approach, we aim to produce a simulation system blueprint that is optimized for local and regional action, including the identification and prioritization of local knowledge gaps.

Essential Qualification: PhD in Geoscience/ Civil Engg/ physics/ Data-science/ Remote-sensing/ Water research/ Environmental sciences.

Desirable requirement: Good knowledge of programming, particularly in Python/MATLAB and/or R/c/c++, additionally, the candidate with good knowledge of hydrological modelling, application of AI based data-assimilation, Bayesian inference will be ideal. Any prior experience in Data assimilation will be a plus.

Duration: One year at first, extendable to 4 more years upon satisfactory progress.

Salary: As per Institute norms with eligible increments. Minimum around 80K per month consolidated for candidates just out of PhD. Experienced candidates are eligible for higher salary.

Other incentives: Work with an international team, International exposure, career guidance and chance to be involved in cutting edge research.

How to apply: Interested candidates are requested to send a cover letter (describing their motivation and interests), their CV/resume along with the name and contact information of two referees to bramha@iisc.ac.in. The last date to apply is 5th August 2026.